Mathematical Physics I (Fall 2026): Homework #4 Solution

Due Nov. 6, 2026 (Fri, 23:00pm)

[0.5 pt each, total 5 pts]

1. Boas Chapter 7, Problem 5.3

• From Eqs.(5.6), (5.9)-(5.10), you can find

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{\pi}\int_{\pi/2}^{\pi} dx = \frac{1}{2},$$

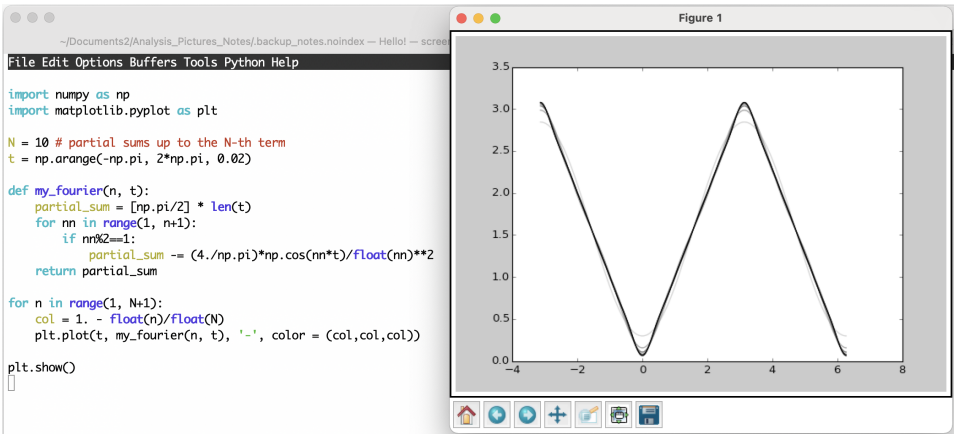
$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\cos nx\, dx = \frac{1}{\pi}\int_{\pi/2}^{\pi} \cos nx\, dx = \frac{1}{\pi}\left[\frac{\sin nx}{n}\right]_{\pi/2}^{\pi} = \begin{cases} 0, & \text{if } n \text{ even}, \\ \frac{(-1)^{\frac{n+1}{2}}}{n\pi}, & \text{if } n \text{ odd}, \end{cases}$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)\sin nx\, dx = \frac{1}{\pi}\int_{\pi/2}^{\pi} \sin nx\, dx = \frac{1}{\pi}\left[-\frac{\cos nx}{n}\right]_{\pi/2}^{\pi} = \begin{cases} \frac{1}{n\pi}, & \text{if } n \text{ odd}, \\ 0, & \text{if } n = 4k, \\ -\frac{2}{n\pi}, & \text{if } n = 4k+2. \end{cases}$$

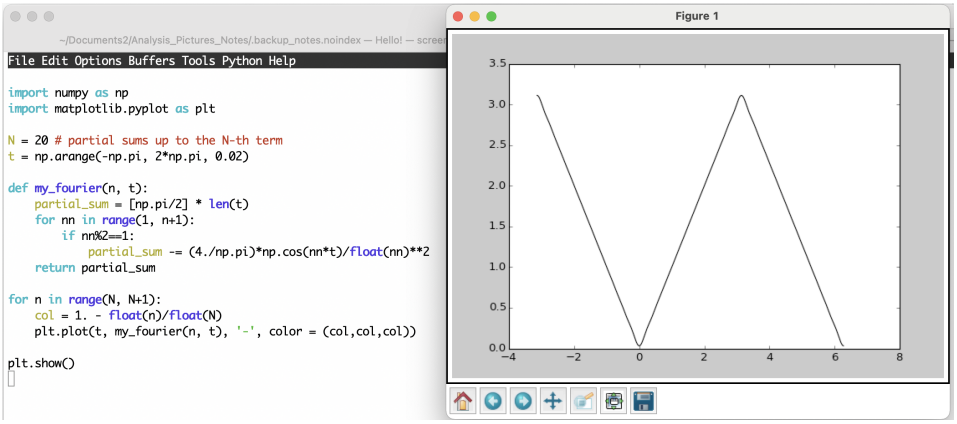
2. Boas Chapter 7, Problem 6.12 (for Problem 5.9)

(Note: For Problem 6.12, you will first have to work out Problem 5.9. You are asked to tackle Problem 5.9 in two ways — first directly find the Fourier coefficients, then verify your result using the answer given in Problem 5.7. Then for Problem 6.12 you may consider a way to automatically draw several partial sums of different n's — e.g., a simple script written in MATLAB or in `python`; however, making such an automated script is not necessary for you to receive a full credit.)

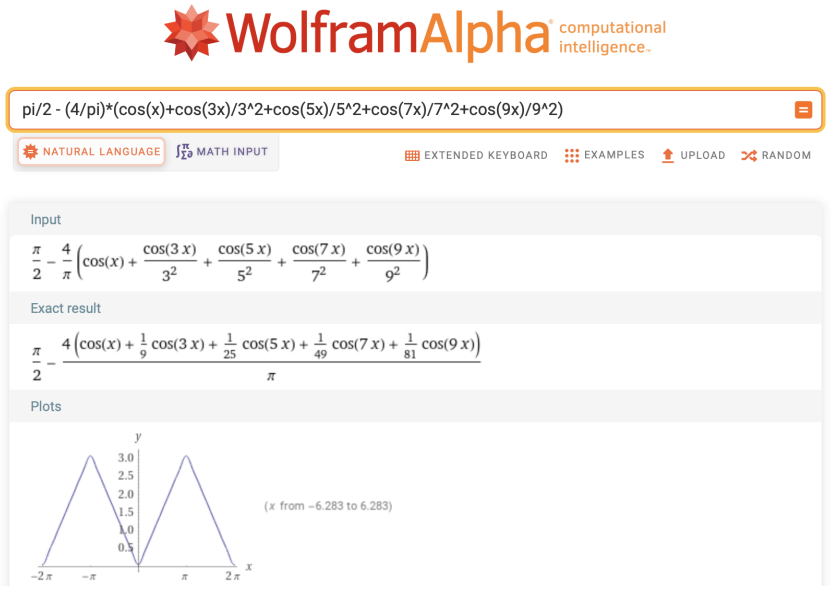
• An example of an automated plotting script ($n = 1$ to 9) is as follows:



By pushing n to 20, a very good approximation of the given $f(x)$ is acquired already:



• You are, however, not required to come up with an automated script like above. You may simply use e.g., WolframAlpha to create multiple plots in a row. For example, the black line ($n = 9$) in the first plot above matches what is seen below:



3. Boas Chapter 7, Problem 7.11

• From Eqs.(7.4) and (7.6) of Boas Chapter 7, you can find

$$c_0 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{1}{2\pi}\int_{0}^{\pi} \sin x\, dx = \frac{1}{\pi},$$
$$c_1 = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-ix}dx = \frac{1}{4i\pi}\int_{0}^{\pi} (e^{ix} - e^{-ix})e^{-ix}dx = \frac{1}{4i} = -\frac{i}{4},$$
$$c_{-1} = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{ix}dx = \frac{1}{4i\pi}\int_{0}^{\pi} (e^{ix} - e^{-ix})\, e^{ix}dx = -\frac{1}{4i} = \frac{i}{4},$$

and for $|n| \geq 2$,

$$c_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)e^{-inx}dx = \frac{1}{2\pi}\int_0^{\pi} \sin x e^{-inx}dx = \frac{1}{4i\pi}\int_0^{\pi}(e^{ix}-e^{-ix})e^{-inx}dx$$
$$= \frac{1}{4i\pi}\left\{\left[\frac{e^{-i(n-1)x}}{-i(n-1)}\right]_0^{\pi} - \left[\frac{e^{-i(n+1)x}}{-i(n+1)}\right]_0^{\pi}\right\} = \frac{1}{4\pi}\left\{\left[\frac{e^{-i(n-1)x}}{(n-1)}\right]_0^{\pi} - \left[\frac{e^{-i(n+1)x}}{(n+1)}\right]_0^{\pi}\right\}$$
$$= \frac{1}{4\pi}\left\{\frac{(-1)^{n-1}-1}{n-1} - \frac{(-1)^{n+1}-1}{n+1}\right\} = \begin{cases} -\frac{1}{\pi}\cdot\frac{1}{n^2-1}, & \text{if } n \text{ even}, \\ 0, & \text{if } n \text{ odd}. \end{cases}$$

Therefore,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx} = \frac{1}{\pi} + \frac{1}{2}\left(\frac{e^{ix}-e^{-ix}}{2i}\right) - \frac{2}{\pi}\sum_{\substack{n\geq 2 \\ n \text{ even}}}^{\infty} \frac{e^{inx}+e^{-inx}}{2(n^2-1)}.$$

4. Boas Chapter 7, Problem 8.15(b)

• From Eq.(8.3), you can find

$$c_0 = \frac{1}{2}\int_{-1}^{1} f(x)dx = \frac{1}{2}\left\{\int_{-1}^{0}(1+2x)dx + \int_0^1 (1-2x)dx\right\} = 0,$$
$$c_n = \frac{1}{2}\int_{-1}^{1} f(x)e^{-in\pi x}dx = \frac{1}{2}\left\{\int_{-1}^{0}(1+2x)e^{-in\pi x}dx + \int_0^1 (1-2x)e^{-in\pi x}dx\right\}$$
$$= \frac{1}{2}\left\{\int_{-1}^{1} e^{-in\pi x}dx - 2\int_0^1 x(e^{in\pi x}+e^{-in\pi x})dx\right\} = -2\int_0^1 x\cos n\pi x\, dx$$
$$= -2\left\{\left[\frac{x\sin n\pi x}{n\pi}\right]_0^1 - \frac{1}{n\pi}\int_0^1 \sin n\pi x\, dx\right\} = \frac{2[1-(-1)^n]}{n^2\pi^2} = \begin{cases} 0, & \text{if } n \text{ even}, \\ \frac{4}{n^2\pi^2}, & \text{if } n \text{ odd}. \end{cases}$$

One can then easily show that

$$\frac{4}{\pi^2}\sum_{\substack{n=-\infty \\ n \text{ odd}}}^{\infty}\frac{e^{in\pi x}}{n^2} = \frac{4}{\pi^2}\sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty}\frac{e^{in\pi x}+e^{-in\pi x}}{n^2} = \frac{8}{\pi^2}\sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty}\frac{\cos n\pi x}{n^2}.$$

5. Boas Chapter 7, Problem 11.8

(Note: For Problem 11.8, you are asked to first work out Problem 9.10.)

• From Eqs.(5.6), (5.9)-(5.10) or (9.5) of Boas Chapter 7, you can find $b_n = 0$,

$$a_0 = \frac{2}{\pi}\int_{-\pi/2}^{\pi/2}|x|dx = \frac{4}{\pi}\int_0^{\pi/2} x dx = \frac{\pi}{2},$$

and

$$a_n = \frac{4}{\pi}\int_0^{\pi/2} x\cos 2nx\, dx = \frac{4}{\pi}\left\{\left[\frac{x\sin 2nx}{2n}\right]_0^{\pi/2} - \frac{1}{2n}\int_0^{\pi/2}\sin 2nx\, dx\right\}$$
$$= \frac{(-1)^n-1}{n^2\pi} = \begin{cases} 0, & \text{if } n \text{ even}, \\ -\frac{2}{n^2\pi}, & \text{if } n \text{ odd}. \end{cases}$$

• Then from Parseval's theorem in Eq.(11.4) of Boas Chapter 7, you find

$$\frac{1}{\pi}\int_{-\pi/2}^{\pi/2}|x|^2\,dx=\frac{\pi^2}{12}=\left(\frac{1}{2}a_0\right)^2+\frac{1}{2}\sum_{n=1}^{\infty}a_n^2=\frac{\pi^2}{16}+\frac{2}{\pi^2}\sum_{\substack{n=1\\ n\,\text{odd}}}^{\infty}\frac{1}{n^4}\quad\rightarrow\quad\sum_{\substack{n=1\\ n\,\text{odd}}}^{\infty}\frac{1}{n^4}=\frac{\pi^4}{96}.$$

6. Boas Chapter 7, Problem 12.24

• (b) From Eq.(12.15) of Boas Chapter 7, you can find

$$\begin{aligned}g_c(\alpha)&=\sqrt{\frac{2}{\pi}}\int_0^{\infty}f(x)\cos\alpha x\,dx=\frac{1}{\sqrt{2\pi}}\int_0^{\infty}e^{-x}(e^{i\alpha x}+e^{-i\alpha x})\,dx\\&=\frac{1}{\sqrt{2\pi}}\left[-\frac{1}{i\alpha-1}+\frac{1}{i\alpha+1}\right]=\sqrt{\frac{2}{\pi}}\cdot\frac{1}{1+\alpha^2}\,,\end{aligned}$$

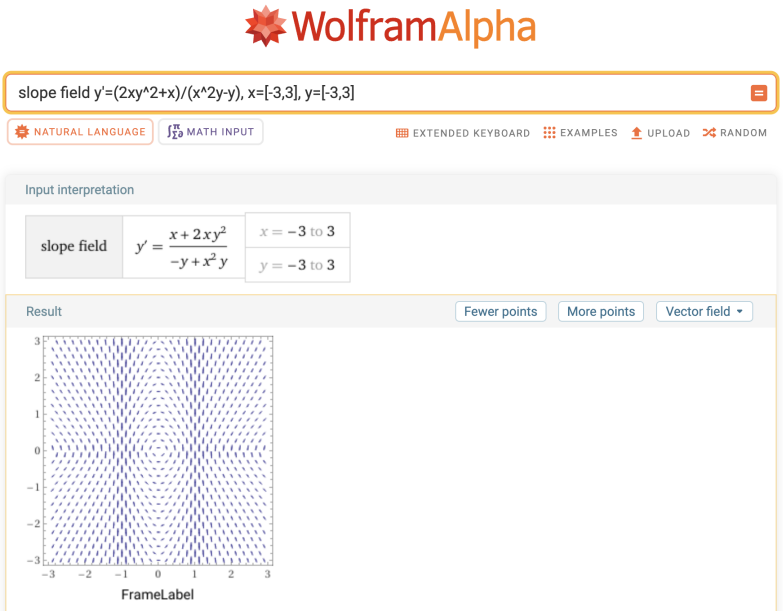
which gives

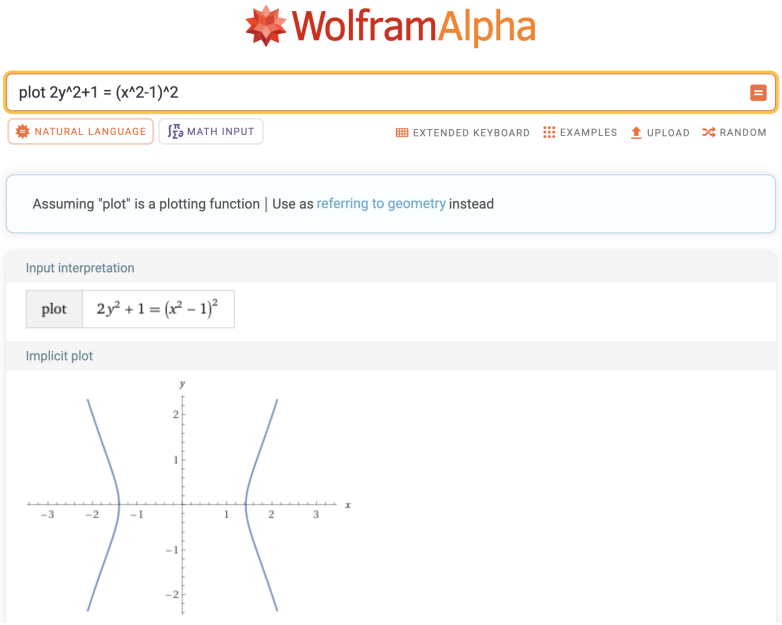
$$f(x)=e^{-|x|}=\sqrt{\frac{2}{\pi}}\int_0^{\infty}g_c(\alpha)\cos\alpha x\,d\alpha=\frac{2}{\pi}\int_0^{\infty}\frac{\cos\alpha x}{1+\alpha^2}\,d\alpha.$$

7. Boas Chapter 8, Problem 2.6

(Note: For Problem 2.6, read the instruction in the textbook carefully; that is, you have been asked to plot a slope field and the solution curve with a computer. To learn more about the "slope field", you may review Boas Chapter 8, Section 1.)

• $(x^2y-y)dy=(2xy^2+x)dx\ \rightarrow\ \frac{ydy}{2y^2+1}=\frac{xdx}{x^2-1}\ \rightarrow\ \frac{1}{4}\ln(2y^2+1)=\frac{1}{2}\ln(x^2-1)+C\ \rightarrow$ $2y^2+1=C'\cdot(x^2-1)^2\ \rightarrow\ (x,y)=(\sqrt{2},0)$ gives $C'=1$.



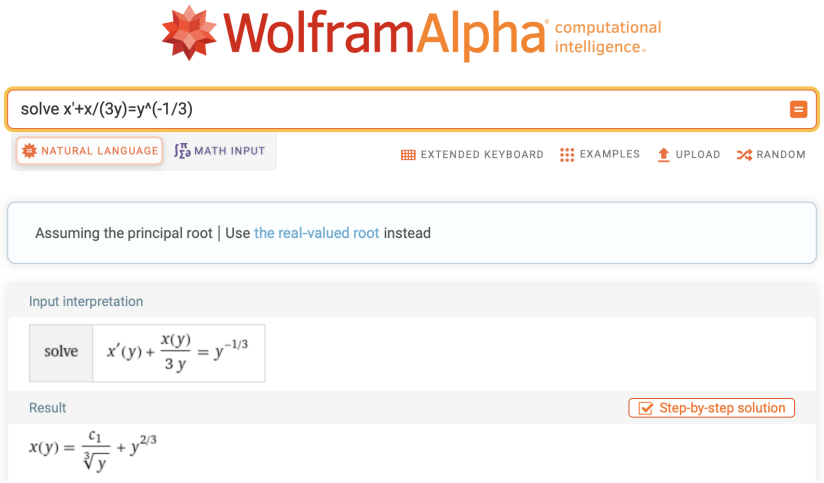


8. Boas Chapter 8, Problem 3.14

(Note: For Problem 3.14, read the instruction and the hint in the textbook carefully.)

• $x' + \frac{1}{3y}x = y^{-\frac{1}{3}} \;\rightarrow\; \frac{dx}{x} = -\frac{dy}{3y} \;\rightarrow\; \ln x = \ln y^{-\frac{1}{3}} + C_1 \;\rightarrow\; x_c = Cy^{-\frac{1}{3}}$. Meanwhile, plugging $x_p = Ay^{\frac{2}{3}}$ in the original equation, you get $A = 1$. Therefore, $x = x_c + x_p = Cy^{-\frac{1}{3}} + y^{\frac{2}{3}}$.

• Alternatively, from Eqs.(3.4) and (3.9), $I = \int P dy = \int \frac{dy}{3y} = \frac{1}{3}\ln y \;\rightarrow\; x = e^{-I}\int Qe^I dy + Ce^{-I} = y^{-\frac{1}{3}}\int y^{-\frac{1}{3}}y^{\frac{1}{3}}dy + Cy^{-\frac{1}{3}} = y^{\frac{2}{3}} + Cy^{-\frac{1}{3}}$.



9. In one of the methods to convert AC voltage into DC voltage, a *full wave diode bridge rectifier* yields the waveform $V(t) = V_0 \left|\sin \omega_0 t\right|$ — i.e., passes the positive peaks of the incoming sine wave but inverts the negative peaks.

(a) After briefly discussing how a full-wave diode bridge rectifier works, expand the rectified sine wave voltage $V(t)$ in an appropriate Fourier series. Show that the lowest (fundamental) oscillation frequency is $2\omega_0$, and explain physically why all odd harmonics are completely absent.

(b) Suppose this rectified voltage signal $V(t)$ is applied across a resistor R. Calculate the total time-averaged electrical power $\overline{P_{\text{total}}}$ dissipated in R directly using the time-domain integration:

$$\overline{P_{\text{total}}} = \frac{1}{T}\int_0^T \frac{[V(t)]^2}{R}\, dt.$$

Then use Parseval's Theorem to express $\overline{P_{\text{total}}}$ as the sum of powers contributed by the DC component and each individual AC harmonic:

$$\overline{P_{\text{total}}} = P_{\text{DC}} + \sum_{n=1}^{\infty} P_n.$$

Calculate the ratio of the power contained in the DC component (P_{DC}) to the total power. Additionally, explain briefly how modern rectifier designs reduce the power remaining in unwanted AC harmonics.

(c) Using the Fourier series found above, prove that:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \frac{1}{7\cdot 9} + \cdots = \sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \frac{1}{2},$$

$$\frac{1}{1\cdot 3} - \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} - \frac{1}{7\cdot 9} + \cdots = \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{4k^2-1} = \frac{\pi-2}{4}, \quad \text{and}$$

$$\frac{1}{1^2\cdot 3^2} + \frac{1}{3^2\cdot 5^2} + \frac{1}{5^2\cdot 7^2} + \frac{1}{7^2\cdot 9^2} + \cdots = \sum_{k=1}^{\infty} \frac{1}{(4k^2-1)^2} = \frac{\pi^2-8}{16}.$$

(Note: To learn more about how various types of rectifiers work in electronic circuits, see e.g., *The Art of Electronics* by Horowitz & Hill. For the last question in (b), you may find the following keywords useful: multi-phase rectifier, smoothing reservoir capacitor, low-pass filter, among others. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)

• (a) From Eqs.(5.6), (5.9)-(5.10) or (9.5) of Boas Chapter 7, you can find $b_n = 0$,

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} V_0 \left|\sin\omega_0 t\right| d(\omega_0 t) = \frac{2V_0}{\pi}\int_0^{\pi} \sin\omega_0 t\, d(\omega_0 t) = \frac{4V_0}{\pi},$$

and

$$a_n = \frac{2V_0}{\pi}\int_0^{\pi} \sin\omega_0 t \cos n\omega_0 t\, d(\omega_0 t) = \frac{2V_0}{\pi}\int_0^{\pi} \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i}\cdot\frac{e^{in\omega_0 t} + e^{-in\omega_0 t}}{2}\, d(\omega_0 t)$$

$$= \begin{cases} \frac{2V_0}{\pi}\left[\frac{1}{2}\left(\frac{1}{n+1} - \frac{1}{n-1}\right) - \frac{1}{2}\left(\frac{1}{n-1} - \frac{1}{n+1}\right)\right] = -\frac{2V_0}{\pi}\frac{2}{n^2-1}, & \text{if } n \text{ even}, \\ 0, & \text{if } n \text{ odd}. \end{cases}$$

Therefore, from Eq.(5.1) of Boas Chapter 7,

$$V(t) = V_0\,|\sin\omega_0 t| = \frac{2V_0}{\pi} - \frac{4V_0}{\pi}\sum_{n\,\text{even}}\frac{\cos n\omega_0 t}{n^2-1},$$

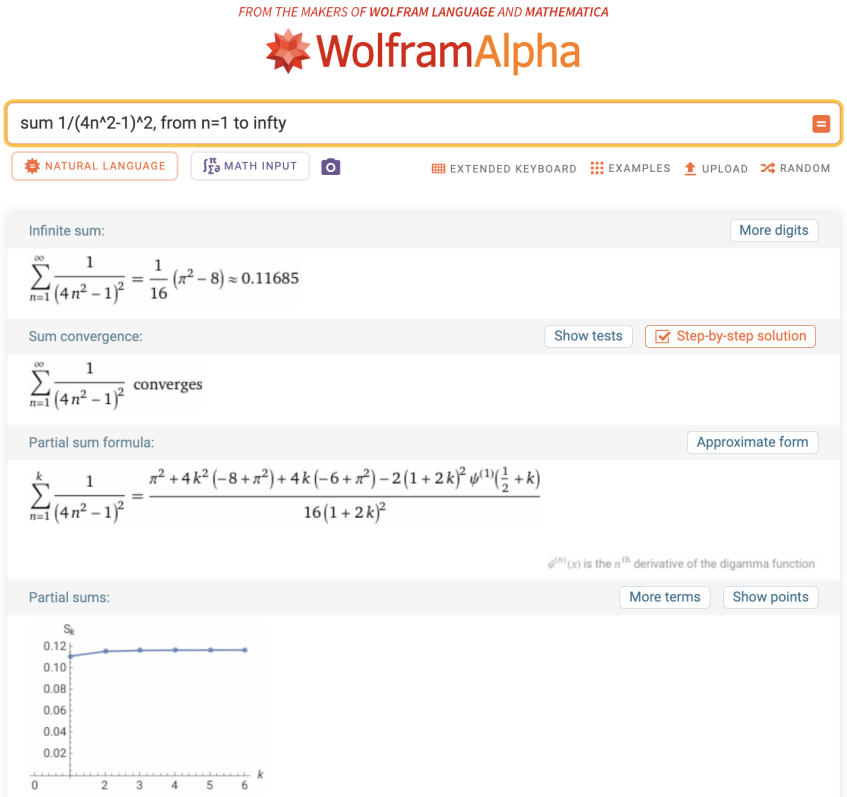
where all odd harmonics are absent because full-wave rectification made the waveform periodic with frequency $2\omega_0$.

• (b) From Eq.(11.4) of Boas Chapter 7, over one period $T = \frac{\pi}{\omega_0}$,

$$\begin{aligned}\overline{P_{\text{total}}} &= \frac{1}{R}\cdot\frac{1}{T}\int_0^T V_0^2|\sin\omega_0 t|^2\,dt = \frac{V_0^2}{2R}\\ &= \frac{1}{R}\left\{\left(\frac{a_0}{2}\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty}a_n^2\right\} = P_{\text{DC}} + \sum_{n=1}^{\infty}P_n = \frac{4V_0^2}{\pi^2 R} + \frac{8V_0^2}{\pi^2 R}\sum_{k=1}^{\infty}\frac{1}{(4k^2-1)^2}.\end{aligned}$$

Thus, the ratio of the DC power to the total power is $\frac{P_{\text{DC}}}{P_{\text{total}}} = \frac{4V_0^2/\pi^2 R}{V_0^2/2R} = \frac{8}{\pi^2} \approx 81\%$.

• (c) You can acquire the desired relations by e.g., plugging $\omega_0 t = 0$ and $\omega_0 t = \frac{\pi}{2}$ into the Fourier cosine series in (a) with $n = 2k$.



10. Let us use the Fourier transform to appreciate the meaning of the Heisenberg uncertainty principle in quantum mechanics.

(a) Imagine an infinite wave train $\sin\omega_0 t$ clipped by shutters to maintain only N cycles of the original waveform:

$$f(t) = \begin{cases} \sin\omega_0 t, & \text{if } |\omega_0 t| < N\pi, \\ 0, & \text{if } |\omega_0 t| > N\pi. \end{cases}$$

Find the amplitude function of the Fourier (exponential) transform, $g(\omega)$. Since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(b) Find the amplitude function of the Fourier sine transform, $g_s(\omega)$. Again, since the prefactor may depend on the exact definition of the transform, do not worry too much about it.

(c) Show that, in the special case of $N = \frac{1}{2}$ and $\omega_0 = 1$, $g_s(\omega)$ becomes

$$g_{s,1}(\omega) = \sqrt{\frac{2}{\pi}}\,\frac{\omega\cos\left(\frac{\omega\pi}{2}\right)}{1-\omega^2}.$$

(d) Now consider the limit of $\omega_0 \gg 1$ and $\omega \approx \omega_0$. Show that $g_s(\omega)$ is approximately equal to

$$g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}}\frac{\sin[(\omega_0-\omega)\frac{N\pi}{\omega_0}]}{\omega_0-\omega}.$$

Sketch or computer plot this function for $N = 1, 3, 5, 10$ and an arbitrary ω_0. Notice that for $N \gg 1$, $g_{s,2}(\omega)$ may be interpreted as proportional to what we later define as the Dirac delta function, $\delta(\omega-\omega_0)$, in Boas Chapter 8, Section 11.

(Note: You do not need to provide a mathematically rigorous proof that $g_{s,2}(\omega)$ indeed becomes proportional to $\delta(\omega-\omega_0)$ as $N \to \infty$. For now, observe the shape of $g_{s,2}(\omega)$ as you vary N, and based on your observation simply argue that $\lim_{\omega\to\omega_0} g_{s,2}(\omega) \to \infty$ as $N \to \infty$.)

(e) Show that the first zeros of $g_{s,2}(\omega)$ from $\omega = \omega_0$ are at $\omega = \omega_0 \pm \Delta\omega = \omega_0 \pm \frac{\omega_0}{N}$. Justify that $\Delta\omega = \frac{\omega_0}{N}$ could be a good measure of the spread (or uncertainty) in frequency of our clipped wave train. Then, establish the inverse relationship between the wave train's pulse length ($N\pi$) and the frequency spread ($\Delta\omega$). Finally, using the relationship along with the assumed *wave nature of matter*, explain the uncertainty principle of quantum mechanics.

(Note: You may want to briefly review the textbooks in quantum mechanics such as Griffiths & Schroeter. Note that the inverse relationship found here is a fundamental property of the finite wave train, and has little to do with any additional *ad hoc* postulates in quantum mechanics. If needed, you must reference your sources appropriately with a proper citation convention, but your answer must still be your own work in your own words. To access the electronic resources — e.g., academic journals — off-campus via SNU library's proxy service, see http://library.snu.ac.kr/using/proxy.)

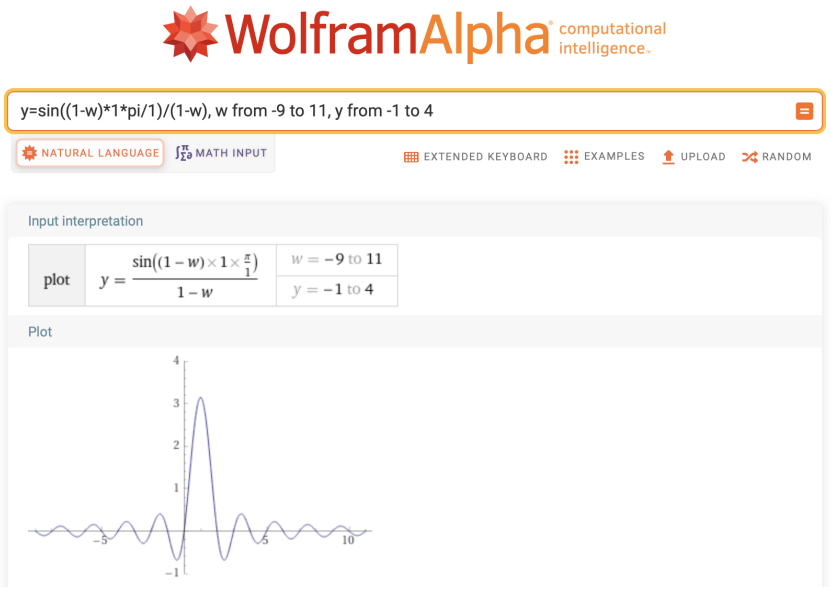
- (a) From Eq.(12.2) of Boas Chapter 7, you find

$$g(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} f(t)e^{-i\omega t}dt = \frac{1}{2\pi}\int_{-\frac{N\pi}{\omega_0}}^{\frac{N\pi}{\omega_0}} \sin\omega_0 t\; e^{-i\omega t}dt = \frac{1}{2\pi}\int_{-\frac{N\pi}{\omega_0}}^{\frac{N\pi}{\omega_0}} \frac{e^{i\omega_0 t} - e^{-i\omega_0 t}}{2i}\cdot e^{-i\omega t}dt$$
$$= \frac{1}{2\pi i}\left[\frac{e^{i(\omega_0-\omega)\frac{N\pi}{\omega_0}} - e^{-i(\omega_0-\omega)\frac{N\pi}{\omega_0}}}{2i(\omega_0-\omega)} - \frac{e^{i(\omega_0+\omega)\frac{N\pi}{\omega_0}} - e^{-i(\omega_0+\omega)\frac{N\pi}{\omega_0}}}{2i(\omega_0+\omega)}\right]$$
$$= \frac{1}{2\pi i}\left[\frac{\sin[(\omega_0-\omega)\frac{N\pi}{\omega_0}]}{\omega_0-\omega} - \frac{\sin[(\omega_0+\omega)\frac{N\pi}{\omega_0}]}{\omega_0+\omega}\right].$$

- (b) From Eq.(12.14) of Boas Chapter 7, you can find

$$g_s(\omega) = \sqrt{\frac{2}{\pi}}\int_0^{\frac{N\pi}{\omega_0}} \sin\omega_0 t\,\sin\omega t\,dt = \frac{1}{\sqrt{2\pi}}\left[\frac{\sin[(\omega_0-\omega)\frac{N\pi}{\omega_0}]}{\omega_0-\omega} - \frac{\sin[(\omega_0+\omega)\frac{N\pi}{\omega_0}]}{\omega_0+\omega}\right].$$

- (d) $\lim_{\omega\to\omega_0} g_{s,2}(\omega) = \frac{1}{\sqrt{2\pi}}\frac{N\pi}{\omega_0} \to \infty$ as $N \to \infty$. Computer plots of $g_{s,2}(\omega)$ for $N = 1$ and 5 can be found below.



WolframAlpha computational intelligence

y=sin((1-w)*5*pi/1)/(1-w), w from -9 to 11, y from -4 to 16

NATURAL LANGUAGE | MATH INPUT | EXTENDED KEYBOARD | EXAMPLES | UPLOAD | RANDOM

Input interpretation

plot	$y = \frac{\sin\left((1-w)\times 5\times\frac{\pi}{1}\right)}{1-w}$	$w = -9$ to 11
		$y = -4$ to 16

Plot

